

# Unsupervised Clustering: HCC Tumor vs Matched Normal Liver

## Team

| Name      | Email            | GitHub                                                                    |
|-----------|------------------|---------------------------------------------------------------------------|
| Junhao Li | lijunhao@ufl.edu | <a href="https://github.com/JunhaoLiXD">https://github.com/JunhaoLiXD</a> |

## Dataset

RNA-seq of 50 Paired Hepatocellular Carcinoma Samples

Source: <https://www.refine.bio/experiments/SRP068976/rna-seq-of-50-paired-hepatocellular-carcinoma?ref=search>

## Scientific Question

Can gene expression profiles distinguish hepatocellular carcinoma (HCC) tumor tissue from matched non-tumor liver tissue?

## GitHub Repository

<https://github.com/JunhaoLiXD/HCC-RNASeq-Classification>

## 1. Load the expression data and metadata

We reused the expression matrix and sample groups processed in Assignment 2 (100 samples: 50 tumor, 50 matched normal). Genes with more than 10 counts in at least 10 samples were kept (23,393 genes), values were  $\log_2(\text{count} + 1)$  transformed, genes were ranked by variance, and the top N genes were z-scored per gene.

## 2. Unsupervised analysis

All main clustering results use the 5,000 most variable genes. As a team of one, two clustering methods were run in Python: **Method 1 – K-means** (scikit-learn, `n_init = 50`, `random_state = 42`) and **Method 2 – hierarchical clustering** (SciPy, average linkage on correlation distance  $1 - \text{Pearson } r$ , tree cut into k clusters with fcluster). Both methods require the number of clusters k to be chosen, so we compared k = 2 to 6 for each and then used k = 2 as the primary k for the gene-number comparison, the heatmap and the statistics. Cluster ids were renumbered 1..k by decreasing size so that versions can be compared.

### 2.1 Method 1: K-means

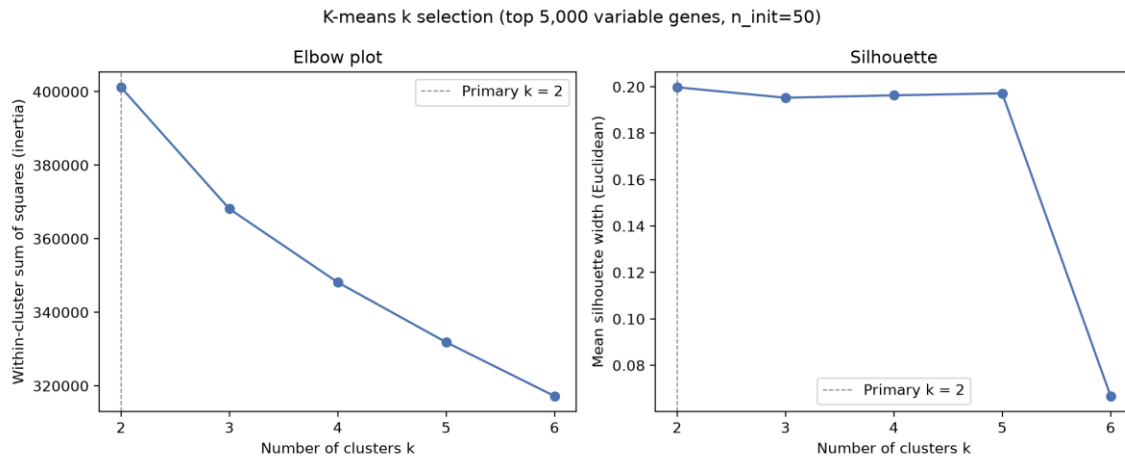


Figure 1. K-means elbow plot (left) and mean silhouette width (right) for k = 2–6, top 5,000 genes.

We ran K-means (`n_init = 50`, `random_state = 42`) on the 100 samples using the 5,000 most variable genes for k = 2 to 6. The left panel shows the within-cluster sum of squares (elbow plot) and the right panel the mean Euclidean silhouette width. Inertia decreases smoothly with no sharp elbow, so the silhouette was used to pick k: it is highest at k = 2 (0.200), stays almost flat through k = 5 (0.195–0.197), and collapses at k = 6 (0.067). We therefore chose k = 2, which also matches the two biological groups; the low absolute silhouette values show that the clusters are real but not sharply separated in 5,000-dimensional space.

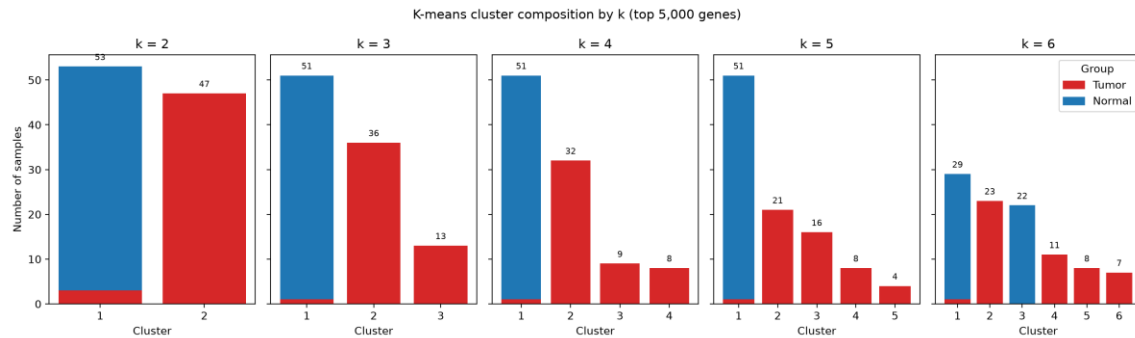


Figure 2. Tumor / Normal composition of each K-means cluster for  $k = 2-6$  (top 5,000 genes); numbers above bars are cluster sizes.

### How many clusters, and how did changing $k$ change the results?

At  $k = 2$  K-means almost reproduces the biological groups: one cluster holds all 50 Normal samples plus 3 Tumor samples and the other holds 47 Tumor samples only. Raising  $k$  from 3 to 5 leaves the Normal cluster intact (50 Normal + 1 Tumor) and only splits the tumors into smaller sub-clusters (36/13 at  $k = 3$ , down to 21/16/8/4 at  $k = 5$ ), suggesting that tumors are more heterogeneous than normal liver. Only at  $k = 6$  does K-means split the Normal samples (29 vs 22), which coincides with the drop in silhouette. One Tumor sample (Patient 39, SRR3129924) is placed with the Normal samples at every  $k$  and with every gene-set size, consistent with the tumor sample that sits near the Normal group in the Assignment 2 PCA.

## 2.2 Method 2: Hierarchical clustering

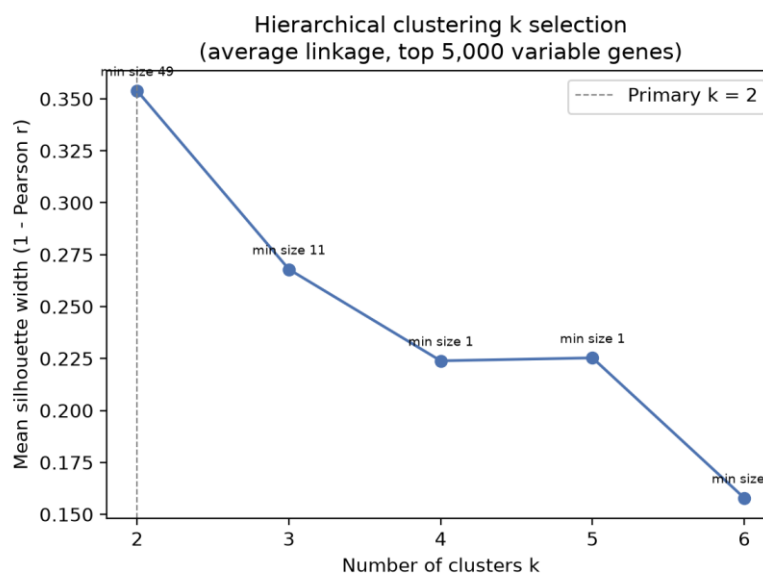


Figure 3. Mean silhouette width (correlation distance) for hierarchical clustering cut at  $k = 2-6$ ; labels give the size of the smallest cluster.

We clustered the samples hierarchically (average linkage, correlation distance  $1 - \text{Pearson } r$ ) on the same z-scored top 5,000 genes and cut the tree into  $k = 2$  to 6 clusters. For each cut we computed the mean silhouette width on the same correlation distance and annotated the size of the smallest cluster. The silhouette is clearly highest at  $k = 2$  (0.354) and falls steadily afterwards (0.268 at  $k = 3$ , about 0.22 at  $k = 4-5$ , 0.158 at  $k = 6$ ), so  $k = 2$  was chosen as the primary  $k$ , in agreement with K-means. From  $k = 4$  onwards the extra clusters are single outlying samples rather than new groups, which is why the curve flattens instead of improving.

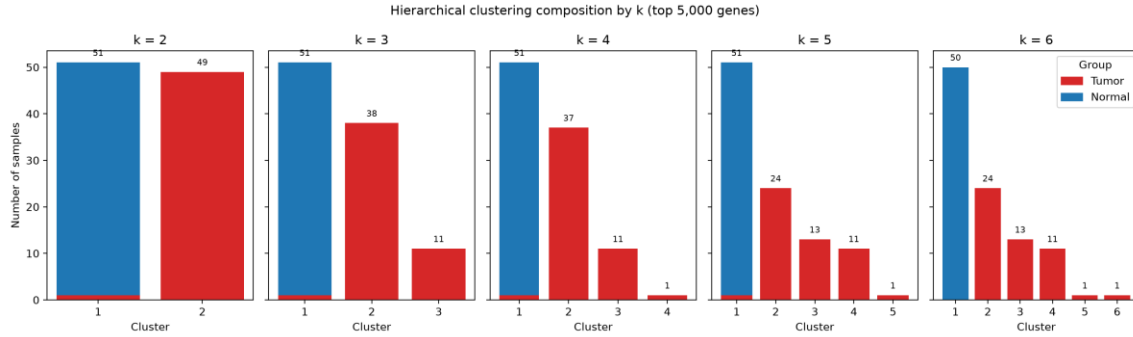


Figure 4. Tumor / Normal composition of each hierarchical cluster for  $k = 2-6$  (top 5,000 genes).

## How many clusters, and how did changing $k$ change the results?

At  $k = 2$  the tree splits into 51 samples (all 50 Normal + 1 Tumor) and 49 Tumor samples, a slightly cleaner split than K-means (3 misplaced tumors). Higher  $k$  only subdivides the Tumor branch (38/11 at  $k = 3$ ), and  $k = 4$  and 5 add a singleton cluster containing one Tumor sample (Patient 36, SRR3129922), an outlier within the tumors. At  $k = 6$  the misplaced Tumor sample (Patient 39) finally leaves the Normal cluster as a second singleton, leaving a pure 50-sample Normal cluster. Average linkage therefore tends to peel off individual outliers as  $k$  grows, whereas K-means prefers more balanced tumor sub-clusters.

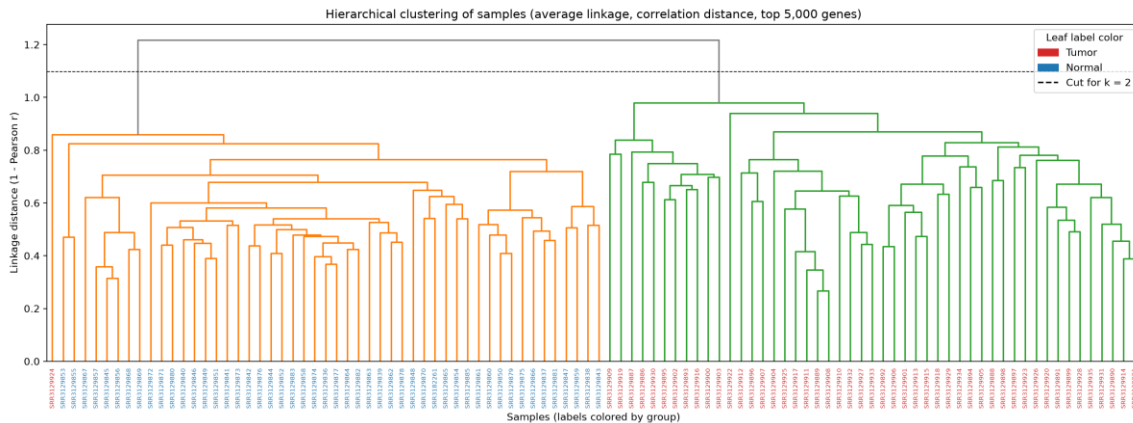


Figure 5. Sample dendrogram (average linkage,  $1 - \text{Pearson } r$ , top 5,000 genes); leaf labels colored by group, dashed line marks the  $k = 2$  cut.

This dendrogram shows the full average-linkage tree of the 100 samples, with leaf labels colored by Tumor / Normal and a dashed line where the tree is cut for  $k = 2$ . The two top-level branches merge at a distance of about 1.2 (the average correlation between Tumor and Normal samples is slightly negative on these z-scored genes), well above any within-branch merge. The Normal branch is compact (merges below about 0.86), while the Tumor branch is deeper and more nested, consistent with the higher heterogeneity of tumors. The only misplaced sample, Patient 39 Tumor, is the last leaf to join the Normal branch, so it is Normal-like but still the most distinct member of that branch.

### 2.3 Number of clusters: comparing $k$

Both methods require  $k$  to be chosen, so we compared cluster membership at every  $k$  from 2 to 6 for both methods on the top 5,000 genes.

**Table 1.** Cluster sizes (Tumor/Normal counts) for  $k = 2-6$ , top 5,000 genes.

| $k$     | K-means clusters                                                         | Hierarchical clusters                                                    |
|---------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| $k = 2$ | 53 (3T/50N), 47 (47T/0N)                                                 | 51 (1T/50N), 49 (49T/0N)                                                 |
| $k = 3$ | 51 (1T/50N), 36 (36T/0N), 13 (13T/0N)                                    | 51 (1T/50N), 38 (38T/0N), 11 (11T/0N)                                    |
| $k = 4$ | 51 (1T/50N), 32 (32T/0N), 9 (9T/0N), 8 (8T/0N)                           | 51 (1T/50N), 37 (37T/0N), 11 (11T/0N), 1 (1T/0N)                         |
| $k = 5$ | 51 (1T/50N), 21 (21T/0N), 16 (16T/0N), 8 (8T/0N), 4 (4T/0N)              | 51 (1T/50N), 24 (24T/0N), 13 (13T/0N), 11 (11T/0N), 1 (1T/0N)            |
| $k = 6$ | 29 (1T/28N), 23 (23T/0N), 22 (0T/22N), 11 (11T/0N), 8 (8T/0N), 7 (7T/0N) | 50 (0T/50N), 24 (24T/0N), 13 (13T/0N), 11 (11T/0N), 1 (1T/0N), 1 (1T/0N) |

Table 1 lists, for each  $k$ , every cluster's size with its Tumor and Normal counts for both methods (full per-cluster tables are in the repository as `kmeans_k_summary.csv` and `hclust_k_summary.csv`). For both methods the Normal samples stay together in one cluster up to  $k = 5$ , and extra clusters only partition the tumors. The methods differ in how they partition the tumors: K-means creates mid-sized tumor sub-groups, while hierarchical clustering isolates single outlying tumors (Patient 36 from  $k = 4$ , Patient 39 at  $k = 6$ ).

### 2.4 Effect of the number of genes

We reran both methods at  $k = 2$  using the 10, 100, 1,000, 5,000 and 10,000 most variable genes (same filtering, log transform and z-scoring).

**Table 2.** Cluster sizes (Tumor/Normal counts) at  $k = 2$  for each number of genes.

| Genes  | K-means clusters         | Hierarchical clusters      |
|--------|--------------------------|----------------------------|
| 10     | 54 (4T/50N), 46 (46T/0N) | 65 (30T/35N), 35 (20T/15N) |
| 100    | 53 (3T/50N), 47 (47T/0N) | 51 (1T/50N), 49 (49T/0N)   |
| 1,000  | 51 (1T/50N), 49 (49T/0N) | 51 (1T/50N), 49 (49T/0N)   |
| 5,000  | 53 (3T/50N), 47 (47T/0N) | 51 (1T/50N), 49 (49T/0N)   |
| 10,000 | 54 (4T/50N), 46 (46T/0N) | 51 (1T/50N), 49 (49T/0N)   |

### **How did the number of genes affect clustering?**

For K-means the number of Tumor samples placed in the Normal-like cluster is 4, 3, 1, 3 and 4 for 10, 100, 1,000, 5,000 and 10,000 genes, so even 10 genes recover the Tumor / Normal split almost perfectly and 1,000 genes give the cleanest split. Hierarchical clustering gives an identical partition (50 Normal + 1 Tumor vs 49 Tumor) for every gene set of 100 or more genes. With only 10 genes, however, hierarchical clustering no longer follows the biological groups (65 vs 35 samples, each a mix of both groups): Pearson correlation computed over just 10 values is noisy, and the 10 most variable genes include XIST and several genes that are not differentially expressed. Overall, both methods are stable once at least 100 genes are used, and correlation-based clustering is the more sensitive of the two to very small gene sets.

### **Chi-squared tests between clustering results.**

To compare the versions formally, we ran a chi-squared test on every pair of gene-number versions within each method (10 pairs per method) and between the two methods at the same number of genes (full results in Table 3, Section 4). All K-means versions agree strongly with each other (ARI 0.77–0.96,  $\text{padj} < 1\text{e-}16$ ), and the hierarchical versions with 100 or more genes are identical (ARI = 1). The only non-significant comparisons all involve the 10-gene hierarchical result (ARI  $\approx 0$ ), which also does not agree with K-means on 10 genes, while K-means and hierarchical clustering agree closely at every gene number from 100 upwards (ARI 0.88–1.00).

### 3. Heatmaps and dendrograms

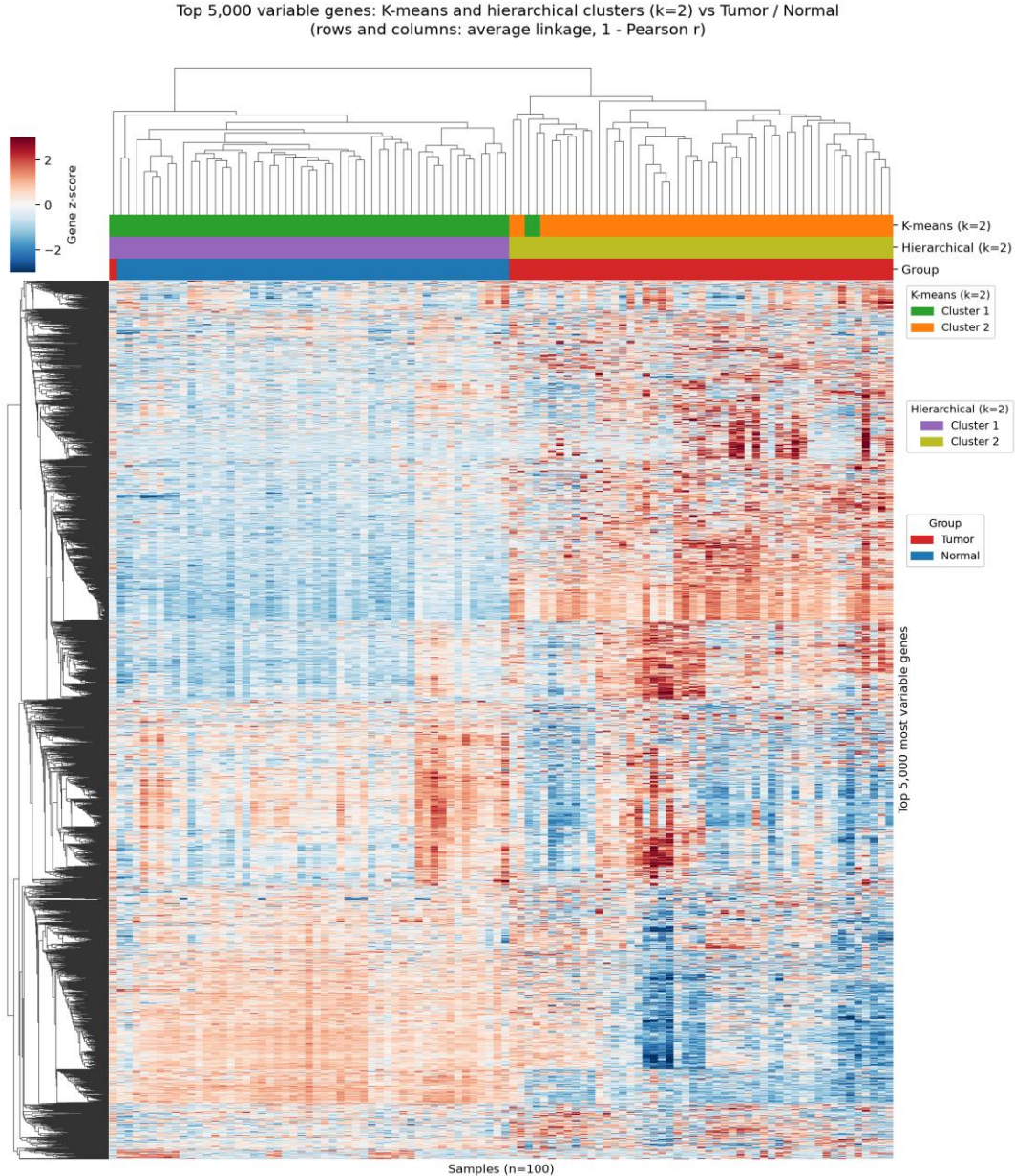


Figure 6. Top 5,000 variable genes (rows, z-scored)  $\times$  100 samples (columns) with row and column dendrograms; annotation bars show K-means ( $k = 2$ ), hierarchical clustering ( $k = 2$ ) and Tumor / Normal.

This heatmap shows the 5,000 most variable genes across all 100 samples, with each gene z-scored across samples and the color scale clipped at  $\pm 3$ . Rows and columns are both clustered with average linkage on correlation distance; the column tree is exactly the one used for hierarchical clustering in Section 2.2 (Figure 5), and the three annotation bars show the K-means clusters, the hierarchical clusters and the Tumor / Normal groups. The column dendrogram splits first into a Normal branch and a Tumor branch, and the hierarchical bar matches the group bar except for one Tumor sample (Patient 39) at the

edge of the Normal branch; K-means additionally places two samples from the Tumor branch with the Normal cluster. The gene blocks show a large set of genes that are uniformly high in normal liver and lower in most tumors, while the tumor-high genes form patchier blocks that are strong in only some tumor subsets, again pointing to tumor heterogeneity.

#### 4. Statistics

We ran 43 chi-squared tests (scipy.stats.chi2\_contingency; Yates continuity correction for 2×2 tables): (i) all 10 pairs of gene-number versions within each method at  $k = 2$ , (ii) K-means vs hierarchical clustering at the same number of genes, and (iii) a test of independence between every clustering result (all  $k$  versions and all gene-number versions) and the Tumor / Normal groups. All p-values were adjusted together with the Benjamini–Hochberg method. As effect sizes we report Cramér's  $V$  (from the uncorrected  $\chi^2$ ) and the adjusted Rand index (ARI).

**Table 3.** Chi-squared test results. KM = K-means, HC = hierarchical clustering; e.g. "HC 1,000g k2" is hierarchical clustering on the top 1,000 genes with  $k = 2$ . padj = BH-adjusted p-value over all tests.

| Comparison     | Result A      | Result B      | $\chi^2$ | dof | p        | padj     | V     | ARI   |
|----------------|---------------|---------------|----------|-----|----------|----------|-------|-------|
| Within K-means | KM 10g k2     | KM 100g k2    | 84.6     | 1   | 3.65e-20 | 5.81e-20 | 0.940 | 0.882 |
| Within K-means | KM 10g k2     | KM 1,000g k2  | 84.9     | 1   | 3.10e-20 | 5.13e-20 | 0.942 | 0.882 |
| Within K-means | KM 10g k2     | KM 5,000g k2  | 77.4     | 1   | 1.42e-18 | 1.69e-18 | 0.900 | 0.808 |
| Within K-means | KM 10g k2     | KM 10,000g k2 | 73.8     | 1   | 8.62e-18 | 1.00e-17 | 0.879 | 0.772 |
| Within K-means | KM 100g k2    | KM 1,000g k2  | 88.5     | 1   | 5.11e-21 | 1.16e-20 | 0.961 | 0.921 |
| Within K-means | KM 100g k2    | KM 5,000g k2  | 80.9     | 1   | 2.33e-19 | 3.04e-19 | 0.920 | 0.845 |
| Within K-means | KM 100g k2    | KM 10,000g k2 | 77.4     | 1   | 1.42e-18 | 1.69e-18 | 0.900 | 0.808 |
| Within K-means | KM 1,000g k2  | KM 5,000g k2  | 88.5     | 1   | 5.11e-21 | 1.16e-20 | 0.961 | 0.921 |
| Within K-means | KM 1,000g k2  | KM 10,000g k2 | 84.9     | 1   | 3.10e-20 | 5.13e-20 | 0.942 | 0.882 |
| Within K-means | KM 5,000g k2  | KM 10,000g k2 | 92.2     | 1   | 7.99e-22 | 2.64e-21 | 0.980 | 0.960 |
| Within HC      | HC 10g k2     | HC 100g k2    | 1.0      | 1   | 3.24e-1  | 3.40e-1  | 0.120 | 0.005 |
| Within HC      | HC 10g k2     | HC 1,000g k2  | 1.0      | 1   | 3.24e-1  | 3.40e-1  | 0.120 | 0.005 |
| Within HC      | HC 10g k2     | HC 5,000g k2  | 1.0      | 1   | 3.24e-1  | 3.40e-1  | 0.120 | 0.005 |
| Within HC      | HC 10g k2     | HC 10,000g k2 | 1.0      | 1   | 3.24e-1  | 3.40e-1  | 0.120 | 0.005 |
| Within HC      | HC 100g k2    | HC 1,000g k2  | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| Within HC      | HC 100g k2    | HC 5,000g k2  | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| Within HC      | HC 100g k2    | HC 10,000g k2 | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| Within HC      | HC 1,000g k2  | HC 5,000g k2  | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| Within HC      | HC 1,000g k2  | HC 10,000g k2 | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| Within HC      | HC 5,000g k2  | HC 10,000g k2 | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| KM vs HC       | KM 10g k2     | HC 10g k2     | 0.3      | 1   | 5.56e-1  | 5.56e-1  | 0.080 | 0.000 |
| KM vs HC       | KM 100g k2    | HC 100g k2    | 88.5     | 1   | 5.11e-21 | 1.16e-20 | 0.961 | 0.921 |
| KM vs HC       | KM 1,000g k2  | HC 1,000g k2  | 96.0     | 1   | 1.13e-22 | 6.92e-22 | 1.000 | 1.000 |
| KM vs HC       | KM 5,000g k2  | HC 5,000g k2  | 88.5     | 1   | 5.11e-21 | 1.16e-20 | 0.961 | 0.921 |
| KM vs HC       | KM 10,000g k2 | HC 10,000g k2 | 84.9     | 1   | 3.10e-20 | 5.13e-20 | 0.942 | 0.882 |
| vs Group       | KM 5,000g k2  | Tumor/Normal  | 84.9     | 1   | 3.07e-20 | 5.13e-20 | 0.942 | 0.882 |
| vs Group       | KM 5,000g k3  | Tumor/Normal  | 96.1     | 2   | 1.37e-21 | 3.93e-21 | 0.980 | 0.770 |

| Comparison | Result A      | Result B     | $\chi^2$ | dof | p        | padj     | V     | ARI   |
|------------|---------------|--------------|----------|-----|----------|----------|-------|-------|
| vs Group   | KM 5,000g k4  | Tumor/Normal | 96.1     | 3   | 1.08e-20 | 2.22e-20 | 0.980 | 0.710 |
| vs Group   | KM 5,000g k5  | Tumor/Normal | 96.1     | 4   | 6.72e-20 | 9.63e-20 | 0.980 | 0.631 |
| vs Group   | KM 5,000g k6  | Tumor/Normal | 96.1     | 5   | 3.44e-19 | 4.35e-19 | 0.980 | 0.385 |
| vs Group   | KM 10g k2     | Tumor/Normal | 81.5     | 1   | 1.73e-19 | 2.33e-19 | 0.923 | 0.845 |
| vs Group   | KM 100g k2    | Tumor/Normal | 84.9     | 1   | 3.07e-20 | 5.13e-20 | 0.942 | 0.882 |
| vs Group   | KM 1,000g k2  | Tumor/Normal | 92.2     | 1   | 7.85e-22 | 2.64e-21 | 0.980 | 0.960 |
| vs Group   | KM 10,000g k2 | Tumor/Normal | 81.5     | 1   | 1.73e-19 | 2.33e-19 | 0.923 | 0.845 |
| vs Group   | HC 5,000g k2  | Tumor/Normal | 92.2     | 1   | 7.85e-22 | 2.64e-21 | 0.980 | 0.960 |
| vs Group   | HC 5,000g k3  | Tumor/Normal | 96.1     | 2   | 1.37e-21 | 3.93e-21 | 0.980 | 0.791 |
| vs Group   | HC 5,000g k4  | Tumor/Normal | 96.1     | 3   | 1.08e-20 | 2.22e-20 | 0.980 | 0.776 |
| vs Group   | HC 5,000g k5  | Tumor/Normal | 96.1     | 4   | 6.72e-20 | 9.63e-20 | 0.980 | 0.649 |
| vs Group   | HC 5,000g k6  | Tumor/Normal | 100.0    | 5   | 5.29e-20 | 8.12e-20 | 1.000 | 0.669 |
| vs Group   | HC 10g k2     | Tumor/Normal | 0.7      | 1   | 4.02e-1  | 4.11e-1  | 0.105 | 0.001 |
| vs Group   | HC 100g k2    | Tumor/Normal | 92.2     | 1   | 7.85e-22 | 2.64e-21 | 0.980 | 0.960 |
| vs Group   | HC 1,000g k2  | Tumor/Normal | 92.2     | 1   | 7.85e-22 | 2.64e-21 | 0.980 | 0.960 |
| vs Group   | HC 10,000g k2 | Tumor/Normal | 92.2     | 1   | 7.85e-22 | 2.64e-21 | 0.980 | 0.960 |

37 of 43 tests are significant ( $\text{padj} < 0.05$ ); the non-significant ones are the six comparisons involving the 10-gene hierarchical result discussed in Section 2.4.

### Does cluster membership correlate with the Assignment 1 groups?

Yes. Every clustering result except hierarchical clustering on 10 genes is strongly associated with the Tumor / Normal groups ( $\text{padj} < 1\text{e-}16$ , Cramér's V 0.92–1.00), while the 10-gene hierarchical result is not ( $\text{padj} = 0.41$ ). For  $k > 2$  Cramér's V stays at about 0.98 while ARI drops (K-means  $0.88 \rightarrow 0.39$ , hierarchical  $0.96 \rightarrow 0.67$ ), because the extra clusters only subdivide the tumors: they remain pure in group label but no longer match a two-group partition. Six tests ( $k \geq 4$  vs group) have expected cell counts below 5 because of small or singleton clusters, so their p-values are approximate.

## Conclusion

Both unsupervised methods recover the Tumor / Normal distinction without using the labels: at  $k = 2$  they misplace only 1–3 of 100 samples, and  $k = 2$  has the highest silhouette for both methods. Larger  $k$  values reveal structure within the tumors but not within normal liver, indicating that HCC tumors are transcriptionally more heterogeneous than matched normal tissue. One tumor sample (Patient 39) clusters with normal liver in every  $k = 2$  result, and the clustering is robust to the number of genes from 100 genes upwards. This answers our scientific question: gene expression profiles alone are sufficient to separate HCC tumor tissue from matched normal liver.